



# Effects of Artificial Intelligence Feedback on Students' Self-Regulated Learning in Higher Education: A Three-Level Meta-Analysis

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## Abstract

As higher education environments become digitized and complex, self-regulated learning (SRL) remains a pivotal cornerstone of student achievement. While artificial intelligence (AI) offers new possibilities for supporting SRL through feedback, its efficacy remains controversial. A three-level meta-analysis of 85 articles ( $k=387$ ;  $N=9,564$ ) published between 2015 and 2025 was conducted. The results demonstrated a significant, medium-sized positive effect of AI feedback on students' SRL ( $g=0.509$ ). Dimension-specific analyses revealed that effects were stronger for the motivational and cognitive dimensions, but comparatively weaker for behavioral and metacognitive outcomes. Notably, the wide 95% prediction interval  $[-0.661, 1.678]$  indicates substantial heterogeneity, suggesting that AI feedback effects are conditional rather than uniformly beneficial. Hierarchical multilevel meta-regression identified feedback source and AI technology type as primary drivers of between-study heterogeneity. AI-led feedback yielded larger effect sizes than human-AI hybrid feedback, and generative AI dialogue systems and intelligent tutoring systems demonstrated stronger effects than non-generative AI dialogue systems and learning analytics systems. Feedback modality and academic subject yielded marginally significant or dimension-specific differences. Additionally, within-category analyses indicated that the effects of generative AI dialogue systems were not significantly moderated across conditions, with the exception of a marginal effect for feedback modality. However, within the learning analytics systems group, the study design demonstrated a significant moderating role. Overall, these findings not only quantify the efficacy of AI feedback on SRL outcomes, but also delineate its boundary conditions, offering empirically grounded implications to instructors, designers, and researchers.

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**Keywords** Artificial intelligence · Feedback · Self-regulated learning · Higher education · Generative AI · Meta-analysis

## Introduction

In increasingly complex and digitalized higher education environments, self-regulated learning (SRL) has become critical to academic success and lifelong learning (Broadbent & Poon, 2015; Panadero, 2017; Pintrich, 2000). The advancement of artificial intelligence (AI) technologies has introduced new possibilities for supporting SRL. AI feedback, with its capacity for personalization, immediacy, and formative guidance, represents a promising alternative to traditional feedback mechanisms (Ba et al., 2025; S.-H. Jin et al., 2023). Despite these advantages, empirical evidence regarding the effectiveness of AI feedback on SRL remains fragmented and controversial. While some research has shown that AI feedback can support SRL by providing timely, personalized, and process-oriented guidance (Chen, 2024; Dever et al., 2022; Ustun et al., 2023), other studies have identified negative consequences, including AI hallucinations affecting cognitive judgments and metacognitive laziness (Fan et al., 2024; Ji et al., 2024). These inconsistencies may stem from moderating factors related to AI feedback, study context, and methodological factors, but a quantitative synthesis remains limited. This study conducted a three-level meta-analysis of empirical research to evaluate: (1) the overall effect of AI feedback on students' SRL in higher education; and (2) potential moderating factors that explain variations in its impact across studies.

## Self-Regulated Learning in Higher Education

Unlike structured K-12 environments, higher education demands a higher degree of autonomy, requiring students to actively construct their learning process rather than passively receive information (Bollas & Bennett, 2026). Based on Pintrich (2000), SRL is conceptualized as an active, constructive process of goal-directed regulation. This framework comprises four key dimensions: cognitive (information processing), metacognitive (monitoring and control), motivational (value beliefs and affect), and behavioral (resource management). Crucially, SRL is viewed not as a fixed trait but as a malleable capacity that develops through targeted external support, whether from instructors or technological systems (Ortubé et al., 2024; Theobald, 2021). In higher education, effective SRL support strikes a balance between external scaffolding and student autonomy, providing appropriate guidance while fostering internal regulation skills (Gupta et al., 2024; Russell et al., 2022).

## Feedback in Self-Regulated Learning

Feedback serves as a vital bridge between students' current performance and their goals, exposing discrepancies that activate cognitive, metacognitive, motivational, and behavioral dimensions of SRL (Butler & Winne, 1995; Hattie & Timperley, 2007). The effectiveness of feedback depends on factors captured in the MISCA

(Message, Implementation, Student, Context, and Agents) model (Panadero & Lipnevich, 2022). Message specifies the content of feedback regarding students' performance. Implementation refers to how feedback is delivered and enacted in instructional practice. Student characteristics reflect individual differences that influence how feedback is perceived, processed, and applied. Context refers to the learning environment, instructional design, and external conditions that ensure feedback is successfully received and utilized. Agents denote the sources of feedback, including teachers, peers, students themselves, and external systems or tasks.

### The Effect of AI Feedback on Students' SRL

AI feedback, defined as information generated by AI systems about students' understanding or performance, aims to bridge the gap between current and desired states (Banihashem et al. 2025a; Lodge et al. 2018). Compared to human feedback, AI feedback offers potential advantages in terms of personalization, immediacy, efficiency, and formative assessment (Letteri & Vittorini, 2025). Through adaptive and targeted feedback, AI can guide students to reflect on their goals, monitor progress, and revise strategies accordingly. When such feedback supports autonomy, enhances competence, and fosters relatedness (Deci & Ryan, 2000; Ryan & Deci, 2000), it fulfills students' basic psychological needs, fostering sustained motivation to self-regulate. However, significant challenges exist, including AI hallucinations that affect cognitive judgments (Ji et al., 2024), metacognitive laziness where students outsource critical thinking (Fan et al., 2024), and potential dependency that reduces academic autonomy. Consequently, the effectiveness of AI feedback on students' SRL remains controversial.

### Previous Reviews and Research Gaps

Existing syntheses have established AI-driven support on learning outcomes. Recent meta-analyses indicate that intelligent feedback (Zheng et al., 2025) and generative AI (GenAI) interventions (J. Wang & Fan, 2025) significantly enhance students' academic performance and higher-order thinking. Heung & Chiu (2025) reported positive associations between GenAI and student engagement. However, these studies prioritize cognitive performance outcomes, leaving the underlying psychological mechanisms, particularly how students regulate their learning with AI, underexplored. Furthermore, although emerging reviews have begun to unpack features of AI feedback, such as the role of generative models in providing multi-dimensional feedback (Ba et al., 2025) and the use of learning analytics in supporting assessment and feedback practices (Sembey et al., 2024), as well as the design of GenAI-supported SRL activities (Xia et al., 2026), these existing reviews remain primarily narrative or design-focused. Crucially, there remains a scarcity of statistical syntheses regarding how AI feedback impacts the dimensions of SRL. Finally, regarding SRL interventions, prior meta-analyses have primarily focused on non-AI contexts (e.g., Dignath et al., 2023; Panadero et al., 2023). Although recent qualitative reviews suggest that AI tools (e.g., chatbots) can support SRL processes like information seeking (Banihashem et al. 2025a; Lan and Zhou 2025), they do not provide meta-analytic evi-

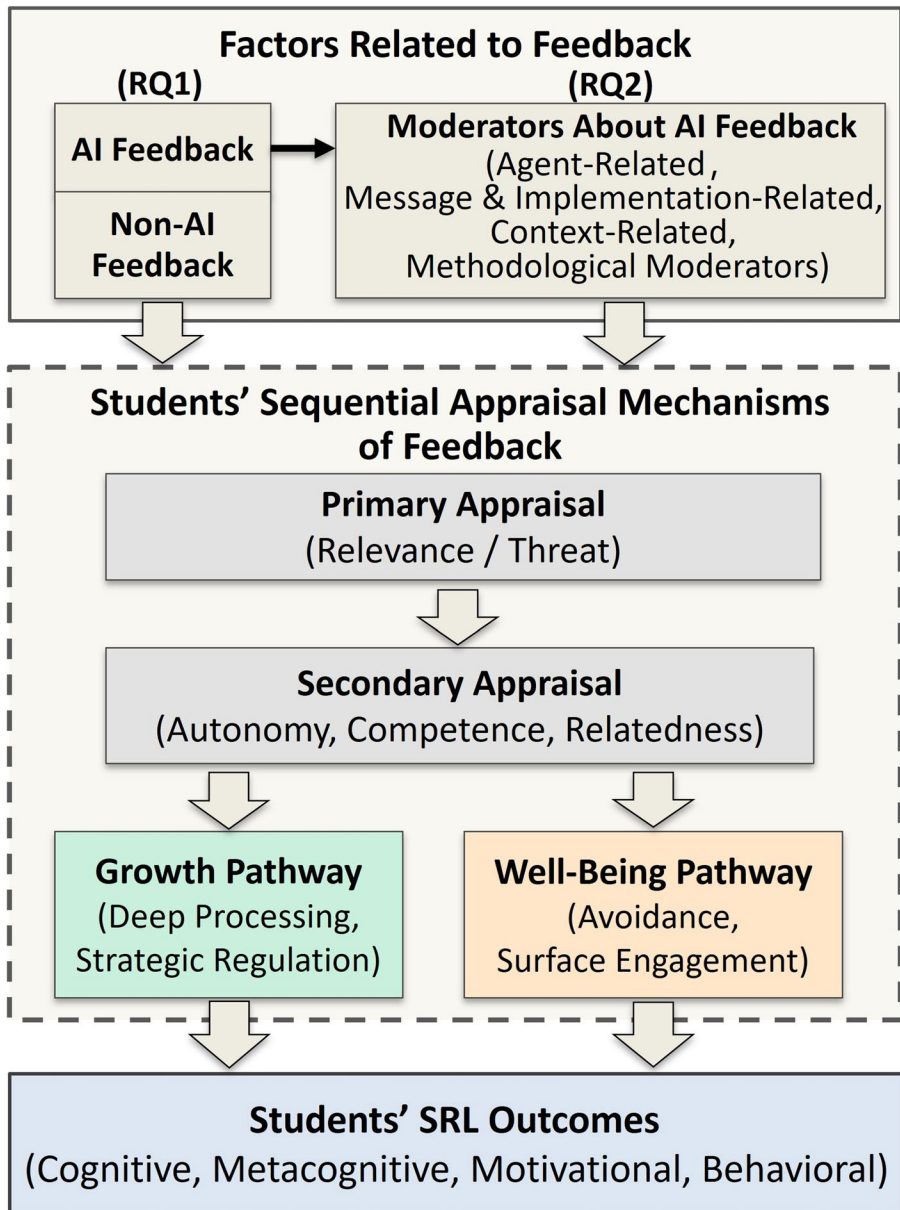
dence. Moreover, while empirical studies show mixed results regarding AI's impact on SRL, a theory-driven, systematic exploration of its boundary conditions remains lacking, particularly concerning how agent characteristics, pedagogical implementations, and contextual factors moderate the effects of AI feedback on students' SRL. Consequently, a meta-analysis is necessary not only to quantitatively clarify the impact of AI feedback on students' SRL outcomes, but also to employ an integrative framework to unpack the moderating factors driving these empirical inconsistencies.

## Conceptual Framework and Research Questions

To synthesize the overall and dimension-specific effects of AI feedback versus non-AI feedback on students' SRL, identify moderators explaining heterogeneous findings, and explain how AI feedback differs from non-AI feedback in influencing SRL outcomes, this study proposes an integrative conceptual framework (Fig. 1).

Guided by the MISCA model (Panadero & Lipnevich, 2022) as a theoretical basis and in light of the requirements of quantitative synthesis, this framework systematically identifies key moderators and classifies them into four categories. First, agent-related moderators (e.g., feedback source, AI technology type) capture the identity and relational characteristics of the feedback provider. Second, given that feedback content and delivery are often intertwined in AI systems (Ba et al., 2025; Deeva et al., 2021), the message and implementation-related moderators (e.g., AI feedback message type, support type, and modality) were considered jointly to represent embedded instructional cues. Third, context-related moderators (e.g., intervention length, academic subject) describe the broader learning conditions. Finally, methodological moderators were included to account for variability arising from differences in study design and SRL measurement.

To provide an integrated theoretical perspective for understanding the mechanisms underlying the differences between AI feedback and non-AI feedback in influencing SRL outcomes, this framework is grounded in Boekaerts' (2011) dual-processing model of self-regulation. Self-determination theory (SDT; Ryan & Deci, 2000) is further incorporated to elaborate the secondary appraisal process, thereby developing the students' sequential appraisal mechanisms of feedback. Drawing on Boekaerts' (2011) model, feedback is conceptualized as an external stimulus that triggers a dynamic appraisal process. Initially, students engage in a fast primary appraisal to evaluate the feedback's personal relevance and potential threat. Following this initial evaluation, students engage in a secondary appraisal to assess their available resources to cope with the task. SDT further informs this interpretation by underscoring the potential role of students' perceived autonomy, competence, and relatedness mainly during the secondary appraisal (Deci & Ryan, 2000; Ryan & Deci, 2000). Students' responses to feedback be interpreted as aligning with different regulatory tendencies, depending on the outcomes of these appraisals. When feedback is perceived to satisfy basic psychological needs, it aligns with a growth pathway defined by deeper processing and strategic regulation (Vansteenkiste et al., 2004). By contrast, the frustration of these needs corresponds to a well-being pathway characterised by self-protective strategies such as avoidance or surface engagement, adopted to minimise cognitive effort or anxiety (Allen & Kizilcec, 2024; Fan et al., 2024;



**Fig. 1** An integrative conceptual framework for examining AI feedback effects on students' SRL. *Note.* The “Moderators About AI Feedback” section is informed by the *MISCA* model (Panadero & Lipnevich, 2022). The “Students' Sequential Appraisal Mechanisms of Feedback” section is grounded in Boekaerts' (2011) dual-processing model of self-regulation, with self-determination theory (SDT; Ryan & Deci, 2000) incorporated to further elaborate secondary appraisal. Students' SRL outcomes are conceptualized across four dimensions based on Pintrich (2000). The dashed-line boxes labeled “Students' Sequential Appraisal Mechanisms of Feedback” represent an integrated theoretical perspective rather than a causal or mediational model

Kluger & DeNisi, 1996). Ultimately, these pathways are reflected in variations in SRL outcomes across cognitive, metacognitive, motivational, and behavioral dimensions (Pintrich, 2000).

Building on this process-oriented perspective, the framework maps the moderator categories to their distinct roles in shaping and contextualizing students' sequential appraisal of feedback. Agent-related moderators are assumed to primarily shape students' initial appraisal of feedback providers, given that the provider's identity and nature inform rapid relational judgments that dictate whether feedback is perceived as a source of relevance or a threat (Henderson et al., 2025; Kluger & DeNisi, 1996; X. Zhao et al., 2025). Following this initial appraisal, message- and implementation-related moderators are expected to be more closely associated with how feedback is interpreted during secondary appraisal. Features such as feedback message, support type, and modality can be conceptualized as cognitive and pedagogical cues linked to students' perceived autonomy, competence, and relatedness (Annamalai & Nasor, 2025; Deci & Ryan, 2000; Hattie & Timperley, 2007). Context-related moderators delineate the broader conditions under which the sequential appraisal processes unfold, whereas methodological moderators capture variation in how such relationships are operationalized across studies.

Guided by the conceptual framework, this study aims to quantitatively analyze the overall and dimension-specific impacts of AI feedback on students' SRL and identify boundary conditions by examining potential moderators. To account for dependency from studies reporting multiple outcomes, this research employs a three-level meta-analysis to maintain statistical independence while maximizing inclusion of empirical data (Cheung, 2019). Specifically, it addresses:

- **RQ1:** Compared to non-AI feedback, what are the overall and dimension-specific effects of AI feedback on students' SRL (i.e., cognitive, metacognitive, motivational, and behavioral dimensions) in higher education?
- **RQ2:** How do research characteristics (i.e., feedback source, AI technology type, AI feedback message type, AI feedback support type, AI feedback modality, intervention length, academic subject, study design, and SRL measurement method) moderate the overall and dimension-specific effects of AI feedback on students' SRL (i.e., cognitive, metacognitive, motivational, and behavioral dimensions) in higher education?

## Potential Moderators and Hypotheses

Guided by the conceptual framework, the following potential moderators were considered, and corresponding hypotheses were proposed.

### 1) Feedback Source.

Feedback may be provided exclusively by AI (AI-led) or jointly with humans (human–AI hybrid) (Banihashem et al., 2025b). While AI offers scalability, empirical evidence suggests that integrating teacher expertise enhances strategic regulation (Chu et al., 2023), whereas fully automated interventions may weaken effort regula-

tion (Huang et al., 2023). From the perspective of the conceptual framework, human involvement may add relatedness and socio-emotional cues that shape how feedback is appraised. This human element buffers the anxiety often associated with opaque systems, increasing the likelihood that feedback is appraised as a supportive challenge rather than a threat (Winstone et al., 2017).

Hypothesis 1(H1): Human–AI hybrid feedback will yield larger effect sizes on SRL than AI-led feedback.

## 2) AI Technology Type.

Based on the characteristics of AI technology, this study categorized AI feedback into four types (Le et al., 2025; Maity & Deroy, 2024; Rodríguez-Ortiz et al., 2025). GenAI dialogue systems feature natural language interaction and high adaptability. By allowing students to negotiate meaning, these systems directly satisfies the need for autonomy, promoting motivational engagement and deep elaboration (Annamalai & Nasor, 2025; Zhang & Liu, 2025). Similarly, intelligent tutoring systems (ITSs) provide structured, step-by-step scaffolding with immediate mastery-based feedback (Létourneau et al., 2025). This precise guidance supports the need for competence, thereby fostering the cognitive dimension of SRL (Ma & Chen, 2025). Learning analytics systems (LASs) visualize performance data (e.g., progress tracking) but often highlight gaps without immediate instructional support (Kitto et al., 2025). This visibility is prone to being appraised as an evaluative threat, leading to anxiety or avoidance rather than adaptive regulation (Chen, 2024). Finally, non-generative AI dialogue systems provide static, verification-based feedback (e.g., grammar corrections) without deep explanation. Students often perceive this as a superficial remedy, opting for coping strategies such as passive acceptance to minimize cognitive load rather than engaging in deep processing (Y. Jin et al., 2025; Schiller et al. 2024).

Hypothesis 2 (H2): GenAI dialogue systems and ITSs will yield significantly larger effect sizes on SRL than learning analytics systems and non-generative AI dialogue systems.

## 3) AI Feedback Message Type.

This study categorized feedback based on content dimensions: cognitive feedback, metacognitive feedback, motivational feedback, and combinations of these dimensions (Cai et al., 2023; Shute, 2008). Although findings on specific feedback vary (Han et al., 2025; Ortega-Ochoa et al., 2024), single-dimensional feedback is often perceived as fragmented. Accordingly, feedback that integrates multiple dimensions may provide a broader set of coping resources: cognitive cues address immediate knowledge gaps, metacognitive guidance supports strategic monitoring, and motivational elements help sustain engagement (Lukešová & Jennings, 2026). In this sense, more comprehensive feedback provides stronger coping potential to support SRL.

Hypothesis 3 (H3): AI feedback that combines multiple dimensions will yield larger effect sizes on SRL than feedback focusing on a single dimension.

#### 4) AI Feedback Support Type.

AI feedback can be categorized into scaffolded, corrective, and directive types based on the level of support provided (Deeva et al., 2021; Serral & Snoeck, 2016). While empirical studies highlight the strategic value of scaffolded feedback (Chu et al., 2023), others note the risks of over-supporting learners (Han et al., 2025) or the limited motivational impact of simple corrective feedback (C.-C. Liu et al. 2025a, b). From the perspective of the conceptual framework, scaffolded feedback may better support SRL because it provides optimal coping resources that guide students to resolve discrepancies without overwhelming their cognitive capacity, thereby fostering a sense of competence. In contrast, directive or corrective feedback may be less effective when it does not sufficiently help students translate feedback into action or resolve the discrepancy identified in the feedback (Zhan & Yan, 2025).

Hypothesis 4 (H4): Scaffolded feedback will yield larger effect sizes on SRL than corrective or directive feedback.

#### 5) AI Feedback Modality.

Based on Swart et al. (2019) and Cai et al. (2023), the feedback form was categorized as written-text only feedback or multichannel feedback. Written-text only feedback is limited to textual modality, while multichannel feedback incorporates additional modalities beyond text, such as images, audio, or video. Debate persists regarding modality: some studies demonstrate the efficacy of written-text only feedback (Hu et al., 2024), whereas others argue that multichannel formats offer superior engagement advantages (Ustun et al., 2023). In the present framework, multichannel feedback may reduce processing demands and enrich the presentation of feedback, which may make it easier for students to interpret and apply (Wen et al., 2025). By contrast, text-only feedback may place greater demands on decoding and integration, especially when feedback content is complex.

Hypothesis 5 (H5): Multichannel feedback will yield larger effect sizes on SRL than written-text only feedback.

#### 6) Intervention Length.

Findings on intervention length are notably inconsistent: while some meta-analyses report that shorter interventions produce stronger effects (Wu & Yu, 2024), others observe no significant moderating impact (X.-M. Wang et al. 2025a, b). Intervention length is proposed to moderate the stability of appraisal. Supporting SRL constitutes a resource-intensive process of internalization (Butler & Winne, 1995). Short-term interventions may trigger temporary engagement due to novelty, but often fail to shift students from a defensive well-being pathway to a stable growth pathway (Grassini,

2023). Long-term interventions allow students to gradually satisfy needs for relatedness and competence, normalizing AI interaction as a supportive partnership (J. Zhang & Derakhshan, 2025).

Hypothesis 6 (H6): Long-term interventions will yield larger effect sizes on SRL than short-term interventions.

## 7) Academic Subject.

Disciplinary differences yield conflicting expectations: while rule-oriented feedback appears optimal for structured disciplines (Poitras & Lajoie, 2017), AI feedback has also shown significant SRL benefits in less structured domains like language learning (Jones et al., 2018). The proposed framework addresses this by focusing on feedback interpretability. In well-structured domains (e.g., engineering), precise, rule-based feedback satisfies competence needs, activating the growth pathway (Donmez, 2024). Conversely, in less structured domains, AI feedback may be vague or inaccurate due to task complexity, risking the frustration of competence and autonomy, which prompts students to adopt the well-being pathway (e.g., ignoring feedback) (Chiu, 2021).

Hypothesis 7 (H7): Studies conducted in well-structured domains will yield larger effect sizes on SRL than those in less structured domains.

## 8) Study Design.

To isolate the true causal impact of AI feedback, a distinction is drawn between randomized controlled trials (RCTs) and quasi-experimental designs (QEDs) (Makhlouf et al., 2024). RCTs serve as the gold standard for internal validity by minimizing selection bias (Braga et al., 2025). In contrast, meta-analytic evidence suggests that QEDs frequently yield inflated effect sizes due to pre-existing group differences or less rigorous control over confounding variables (Li et al., 2023). QEDs often lack random assignment, potentially including students with pre-existing high motivation or self-regulation skills (selection bias). In contrast, RCTs eliminate these confounding variables, providing a more conservative estimate of the AI's true causal impact (Kraft, 2023). Therefore, QEDs are expected to yield inflated effects relative to the more rigorous controls employed in RCTs.

Hypothesis 8 (H8): Quasi-experimental designs will yield larger effect sizes on SRL than randomized controlled trials.

## 9) SRL Measurement Method.

Given that think-aloud protocols typically lack the statistical data required for meta-analysis, the focus of this paper is on the divergence between self-report questionnaires and behavioral measurements. While some research demonstrates convergence between these approaches (Cheng et al., 2025), others highlight significant discrep-

ancies due to the subjective nature of self-reports (Gu & Yan, 2025). Specifically, susceptible to social desirability bias, students often report their intended adherence to the growth pathway rather than actual strategies, leading to inflated scores. In contrast, behavioral measurements (e.g., log traces) objectively record actual operations and are more sensitive to detecting the surface-level engagement characteristic of the well-being pathway (Ye & Pennisi, 2022). Consequently, subjective measures are expected to overestimate SRL improvements relative to objective traces.

Hypothesis 9 (H9): Self-report measurements will yield larger effect sizes on SRL than behavioral measurements.

## Method

### Literature Search

Following the PRISMA guidelines (Page et al., 2021), we conducted a systematic search across four databases: EBSCOhost (APA PsycArticles, APA PsycInfo, Education Source, ERIC, Teacher Reference Center, Education Research Complete, and Psychology and Behavioral Sciences Collection), Web of Science (Core Collection), Scopus, and ProQuest (Coronavirus Research Database, Education Database, Education Research Index, Publicly Available Content Database). These databases were selected to ensure broad coverage of educational psychology, educational technology, and higher education literature, while Google Scholar was included to identify grey literature and minimize publication bias (i.e., the “file drawer effect”; Rosenthal, 1979). The search strategy was structured around four key conceptual blocks aligned with our research questions: (1) Artificial Intelligence (e.g., “intelligent tutoring system\*”, “generative AI”, “large language model\*”, “chatbot\*”, “educational data mining”), (2) Feedback (e.g., “scaffold\*”, “formative assessment”, “prompt”), (3) Self-Regulated Learning (e.g., “metacognit\*”, “self-efficacy”, “time management”, “learning strateg\*”), and (4) Higher Education (e.g., “undergraduate\*”, “university”, “post-secondary”). The boolean logic employed was: [AI terms] AND [Feedback terms] AND [SRL terms] AND [Higher Education terms]. The search was limited to studies published between January 1, 2015, and December 12, 2025. This process yielded a total of 6,933 records before deduplication (EBSCOhost: 2,108; Web of Science: 1,123; Scopus: 2,548; ProQuest: 1,154). The full search strings and detailed parameters for each database are provided in Supplementary Material A. The screening process followed the PRISMA flow (Fig. 2).

### Inclusion and Exclusion Criteria

From the initial identification of 6,933 records, duplicates were removed, resulting in 2,072 unique records. These were screened by title and abstract, followed by a full-text review of 160 articles using predefined criteria outlined in Table B1 (Supplementary Material B). To be included, studies had to target higher education students and employ an experimental or quasi-experimental design. The intervention must

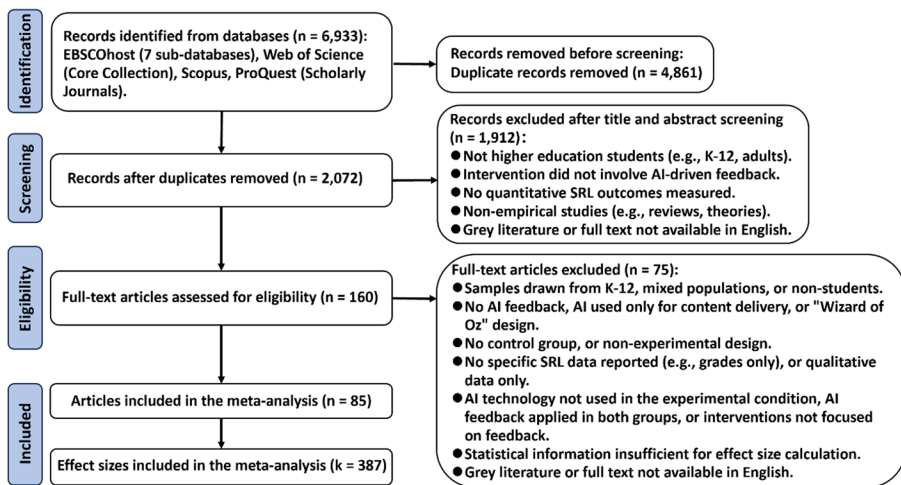


Fig. 2 PRISMA flowchart for including studies to review

have explicitly utilized AI-driven feedback (e.g., generative AI dialogue systems, non-generative AI dialogue systems, intelligent tutoring systems, learning analytics systems) compared against a control group receiving non-AI feedback (e.g., teacher or peer feedback). Regarding outcomes, studies were required to report quantitative SRL measures, either as a composite score or dimensions (cognitive, metacognitive, motivational, or behavioral), utilizing standardized instruments (e.g., MSLQ) or behavioral trace data. Two independent coders cross-coded 50% of the studies to ensure reliability, achieving high inter-rater agreement (Cohen's kappa=0.76–0.93 for title/abstract screening; Cohen's kappa=0.84–0.91 for full-text review). Disagreements were resolved through discussion with a third researcher. The selection process resulted in 85 included articles; see Supplementary Material H for the comprehensive list of studies. Unlike traditional meta-analyses that select a single effect size per study, this study employed a three-level meta-analytic approach. Therefore, multiple effect sizes were extracted from single studies (e.g., reporting on multiple SRL dimensions) to maximize data utilization while statistically accounting for the dependency of effect sizes nested within studies. This resulted in a final dataset of 387 effect sizes.

## Coding and Data Extraction Procedures

Data extraction and coding were systematically performed for all 85 studies meeting the inclusion criteria. To ensure the reliability and rigor of the coding process, a comprehensive codebook was developed and pilot-tested on a random subset of 10 articles, allowing for the refinement of operational definitions prior to full analysis. Two trained researchers independently coded the included articles. Inter-rater reliability was assessed using Cohen's kappa, yielding a coefficient of 0.83, which indicates a high level of agreement (Landis & Koch, 1977). Any discrepancies arising during the coding process were resolved through discussion with a third senior researcher until

full consensus was reached. To enhance transparency and replicability, detailed coding documentation is provided in the appendices. Table C1 (Supplementary Material C) presents the coding framework for SRL and its dimensions, including operational definitions and key indicators grounded in Pintrich's (2000) model. Table D1 (Supplementary Material D) summarizes all moderating factors examined in the meta-analysis, together with their subcategories, descriptions, and illustrative examples. In addition, Table G1 (Supplementary Material G) reports the characteristics of the studies included in the meta-analysis, including author(s) and publication year, SRL dimension, sample size, and the complete set of coded moderator variables.

## Heterogeneity Analysis

Heterogeneity for the overall model was significant,  $Q_{(386)} = 2951.91, p < .001$ . Of the total variation, the sampling variance (Level 1) was 9.90%, the within-study variance (Level 2) was 51.61%, and the between-study variance (Level 3) was 38.49%. The log-likelihood ratio tests (LRT) showed that the between-study variance (Level 3) was significant,  $\chi^2 = 72.95, p < .001$ , suggesting the three-level model was better than the two-level model (Level 3 was set to 0). Similarly, the within-study variance (Level 2) was significant,  $\chi^2 = 611.95, p < .001$ , suggesting the three-level model was better than the two-level model where Level 2 was set to 0. These results justify the use of a three-level model to account for the nested nature of the effect sizes (i.e., multiple effect sizes within studies).

## Statistical Analytic Strategies

The data analysis process adhered to established guidelines for multilevel meta-analyses (Cui et al., 2024; Harrer et al., 2021, Chap. 10; Schindler & Richter, 2023). Effect sizes (Hedges'  $g$ ) and sampling variances were initially computed using Comprehensive Meta-Analysis (CMA) Version 3.7 (Borenstein et al., 2013). Consistent with Cohen (2013),  $g$  values of 0.20, 0.50, and 0.80 were interpreted as small, medium, and large effects, respectively. Subsequent statistical analyses were conducted using the metafor package (Viechtbauer, 2010) in R Version 4.5.2 (R Core Team, 2025). Detailed formulas for effect size conversion and statistical analysis are provided in the Supplementary Materials (Supplementary Material E).

## Quality Assessment

Two independent coders assessed full-text articles using Kmet et al.'s (2004) quality assessment criteria, achieving high inter-rater reliability ( $\kappa = 0.87$ ). The quality assessment items are listed in Supplementary Material X. Disagreements were resolved through discussion, with a third researcher consulted when consensus could not be reached. The 14-item checklist evaluated research design consistency, methodological appropriateness, results precision, and conclusion accuracy using a 3-point scale (2 = "yes," 1 = "partial," 0 = "no"). Items irrelevant to specific factors were marked "N/A" and were excluded from cumulative score calculations.

**Table 1** The overall and dimension-specific effects of AI feedback on SRL

| SRL Outcomes  | k   | n  | Hedges' g | 95% CI         | 95% PI          | p      | Q       |
|---------------|-----|----|-----------|----------------|-----------------|--------|---------|
| Overall SRL   | 387 | 85 | 0.509     | [0.404, 0.613] | [-0.661, 1.678] | <0.001 | 2951.91 |
| Cognitive     | 61  | 28 | 0.478     | [0.260, 0.696] | [-1.075, 2.031] | <0.001 | 673.71  |
| Metacognitive | 95  | 38 | 0.362     | [0.165, 0.559] | [-0.858, 1.582] | <0.001 | 562.61  |
| Motivational  | 105 | 41 | 0.613     | [0.433, 0.794] | [-0.598, 1.824] | <0.001 | 638.82  |
| Behavioral    | 98  | 32 | 0.367     | [0.210, 0.523] | [-0.542, 1.275] | <0.001 | 613.24  |

Note: *k* represents the number of effect sizes, *n* represents the number of studies, *CI* confidence interval, *PI* prediction interval

## Results

### Overall and Dimension-Specific Effects of AI feedback on Students' SRL in Higher Education

To assess the overall efficacy of AI feedback on higher education students' SRL, we synthesized data from 85 articles published between 2015 and 2025 ( $k=387$  effect sizes,  $N=9,564$  students). A three-level random-effects model revealed a statistically significant, medium positive effect of AI feedback on students' SRL ( $g=0.509$ ,  $p < .001$ , 95% CI [0.404, 0.613]). Although the overall effect was robust, the 95% prediction interval (PI) [-0.661, 1.678] and the significant  $Q$ -statistic ( $Q=2951.91$ ,  $p < .001$ ) indicated substantial heterogeneity across the included studies (forest plots are presented in Supplementary Material J, Figs. J3–J7). Subsequent analyses examined the effects of AI feedback on specific SRL dimensions (see Table 1). The results demonstrated that AI feedback yielded significant positive effects across all four dimensions ( $p < .001$ ). Notably, the motivational dimension exhibited the largest effect size ( $g=0.613$ ), followed by the cognitive ( $g=0.478$ ), behavioral ( $g=0.367$ ), and metacognitive ( $g=0.362$ ) dimensions.

### Sensitivity Analyses

To assess whether influential cases distorted the results of the meta-analysis, an influence analysis was conducted based on standardized residuals. The analysis identified 4 outlying effect sizes with absolute  $z$ -scores greater than 3. Results of the influence diagnostics are presented, with influential cases highlighted in red (see Fig. J1 in Supplementary Material J). To evaluate the impact of outliers on the overall effect size, an additional meta-analysis was conducted excluding these cases. The three-level random model showed that the overall effect size remained significant and stable,  $g=0.523$ , 95% CI [0.416, 0.630],  $p < .001$ . Additionally, to assess robustness against study design, a sensitivity analysis restricted to RCTs was conducted. In total, 159 effect sizes from RCTs were synthesized. Results showed that the pooled effect size for overall SRL remained significant,  $g=0.525$ , 95% CI [0.383, 0.667],  $p < .001$ . This result is highly consistent with the main model ( $g=0.509$ ).

## Single Moderator Analysis

To explore the potential sources of heterogeneity, moderator analyses were conducted across the overall SRL and its four dimensions using a mixed-effects model. The results are presented below. As shown in Table 11 (Supplementary Material I), the moderator analysis for the overall SRL effect identified feedback source as a statistically significant moderator ( $F(1, 385)=4.81, p=.029$ ). Additionally, two moderators showed marginally significant effects. The AI technology type was marginally significant ( $F(3, 383)=2.58, p=.054$ ). Feedback modality was also a marginally significant moderator ( $F(1, 385)=3.82, p=.051$ ).

Further moderator analyses were conducted for each SRL dimension, including cognitive, metacognitive, motivational, and behavioral components. Regarding the cognitive dimension (see Table 12 in Supplementary Material I), only AI feedback modality was identified as a marginally significant moderator ( $F(1, 59)=2.86, p=.096$ ). For the metacognitive dimension (see Table 13 in Supplementary Material I), three variables exhibited marginal significance: feedback source ( $F(1, 93)=3.32, p=.072$ ), AI feedback support type ( $F(2, 92)=2.87, p=.062$ ), and academic subject ( $F(5, 89)=2.30, p=.051$ ). Regarding the motivational dimension (see Table 14 in Supplementary Material I), the analysis revealed no statistically significant or marginally significant moderators ( $p>.10$ ). In terms of the behavioral dimension (see Table 15 in Supplementary Material I), study design was identified as a statistically significant moderator ( $F(1, 96)=4.47, p=.037$ ).

### 1) Feedback Source.

Feedback source significantly moderated the effect of AI feedback on students' SRL in higher education. Specifically, AI-led feedback ( $g=0.56$ , 95% CI [0.45, 0.67]) yielded a significantly larger effect size compared to human-AI hybrid feedback ( $g=0.28$ , 95% CI [0.05, 0.51]). This finding stands in direct contrast to hypothesis 1, which predicted that human involvement would enhance effectiveness. Regarding the dimensions, this variable showed a marginally significant moderating effect on the metacognitive dimension. Consistent with the overall trend, AI-led feedback ( $g=0.45$ ) outperformed hybrid feedback ( $g=0.01$ ) in promoting metacognitive outcomes.

### 2) AI Technology Type.

AI technology type was identified as a marginally significant moderator for the overall SRL effect ( $p=.054$ ). Specifically, intelligent tutoring systems ( $g=0.71$ , 95% CI [0.47, 0.94]) and generative AI dialogue systems ( $g=0.58$ , 95% CI [0.42, 0.74]) tended to yield larger effect sizes compared to non-generative AI dialogue systems ( $g=0.39$ , 95% CI [0.13, 0.66]) and learning analytics systems ( $g=0.32$ , 95% CI [0.12, 0.51]). These results support hypothesis 2. However, regarding the dimensions, this variable did not show significant or marginally significant moderating effects for any specific SRL dimension ( $p>.05$ ).

### 3) AI Feedback Message Type.

Despite the lack of statistical significance in the omnibus test, descriptive comparisons suggested that combinatorial feedback types yielded numerically larger effect sizes than single-component feedback. Specifically, metacognitive+motivational feedback ( $g=0.67$ , 95% CI [0.23, 1.11]) and cognitive+metacognitive+motivational feedback ( $g=0.63$ , 95% CI [0.30, 0.95]) tended to produce larger effect sizes compared to motivational feedback alone ( $g=0.32$ ) or cognitive feedback ( $g=0.48$ ). These results offer partial support for hypothesis 3 regarding the value of multidimensional feedback. For the dimensions, although the omnibus tests were not statistically significant, descriptive data highlighted dimension-specific strengths. For instance, in the motivational dimension, metacognitive+motivational feedback ( $g=1.10$ ,  $k=4$ ) demonstrated a particularly strong effect, followed by generative AI dialogue systems utilizing cognitive+metacognitive+motivational feedback ( $g=0.71$ ,  $k=10$ ). Similarly, in the behavioral dimension, cognitive+metacognitive+motivational feedback showed the largest effect size ( $g=0.83$ ,  $k=6$ ), suggesting that comprehensive feedback combining three support types may be most effective. In the cognitive dimension, this comprehensive combination also maintained a high effect size ( $g=0.74$ ,  $k=6$ ).

### 4) AI Feedback Support Type.

For the overall SRL outcomes, AI feedback support type was not a significant moderator ( $p = .527$ ). Descriptive comparisons indicated that directive feedback ( $g=0.56$ , 95% CI [0.38, 0.74]) and scaffolded feedback ( $g=0.51$ , 95% CI [0.37, 0.65]) tended to produce larger effect sizes than corrective feedback ( $g=0.35$ ). This implies that hypothesis 4 (favoring scaffolding) is partially supported. However, it emerged as a marginally significant moderator for the metacognitive dimension ( $p = .062$ ). In this dimension, scaffolded feedback ( $g=0.42$ , 95% CI [0.19, 0.65]) and directive feedback ( $g=0.39$ , 95% CI [0.04, 0.74]) demonstrated positive effects, whereas corrective feedback showed a negative trend ( $g = -0.69$ ), although the latter was based on a small number of effect sizes ( $k=2$ ) and should be interpreted with caution.

### 5) AI Feedback Modality.

Feedback modality was a marginally significant moderator for the overall SRL effect,  $F(1,385)=3.82$ ,  $p = .051$ . AI feedback delivered through multichannel modalities ( $g=0.66$ , 95% CI [0.48, 0.85]) tended to range higher than written-text only feedback ( $g=0.44$ , 95% CI [0.32, 0.56]). This marginally significant finding supports hypothesis 5. This trend was particularly evident in the cognitive dimension, where the moderator effect approached significance ( $F(1,59)=2.86$ ,  $p = .096$ ), with multichannel feedback ( $g=0.73$ ) showing a distinct advantage over text-only feedback ( $g=0.35$ ). The differences were not significant for the metacognitive ( $p = .123$ ), motivational ( $p = .881$ ), or behavioral ( $p = .221$ ) dimensions.

## 6) Intervention Length.

Intervention length did not significantly moderate the overall effectiveness of AI feedback on SRL ( $p=.126$ ). Although descriptive statistics suggested that interventions lasting 6–10 weeks yielded the highest effect size ( $g=0.69$ ), the differences between groups (1 session, 1–6 weeks, 6–10 weeks, and > 10 weeks) were not statistically significant. Therefore, hypothesis 6, which posited a linear benefit of longer interventions, was rejected. This non-significant trend was consistent across all four SRL dimensions ( $p > .05$ ).

## 7) Academic Subject.

The analysis showed that the academic subject was not a significant moderator for the overall SRL effect ( $p = .719$ ). Thus, hypothesis 7 was not supported for overall SRL outcomes. However, it was identified as a marginally significant moderator for the meta-cognitive dimension ( $p = .051$ ). In this dimension, students in natural sciences ( $g=0.89$ , 95% CI [0.44, 1.34]) and interdisciplinary fields ( $g=0.81$ ) appeared to benefit more from AI feedback compared to those in social sciences ( $g=0.06$ ) and languages ( $g=0.09$ ).

## 8) Study Design.

For the overall SRL effect, study design was not a statistically significant moderator ( $p = .667$ ). However, a statistically significant moderating effect was found in the behavioral dimension ( $p = .037$ ). Specifically, RCTs reported a significantly larger effect size ( $g=0.52$ , 95% CI [0.31, 0.73]) compared to QEDs ( $g=0.21$ , 95% CI [0.00, 0.41]), indicating that rigorous experimental conditions might capture stronger behavioral changes induced by AI feedback. This finding contradicts hypothesis 8, which predicted inflated effects in QEDs.

## 9) SRL Measurement Method.

Finally, the method used to measure SRL (self-report questionnaires vs. behavior during an academic task) did not significantly moderate the overall effects of AI feedback ( $p = .87$ ). Consequently, hypothesis 9 was rejected. This lack of significant moderation was consistent across all dimensions ( $p > .05$ ), suggesting that AI feedback is effective in promoting SRL regardless of whether the outcome is measured through self-report questionnaires or behavior during an academic task.

## Multiple Moderator Analysis: Meta-Regression Models

Furthermore, to systematically examine the unique contribution of different levels of moderators while controlling for collinearity, a hierarchical multilevel meta-regression analysis was performed. Guided by the conceptual framework, moderators were entered sequentially according to their distinct roles in shaping and contextualizing students' sequential appraisal mechanisms of feedback. A hierarchical multilevel meta-regression comprising five models was conducted (Models 0–4), specified as three-level random-effects models. Model 0 was the unconditional model. Model 1

entered the context-related moderators (intervention length, academic subject). Model 2 added agent-related moderators (feedback source, AI technology type). Model 3 further incorporated message- and implementation-related moderators (AI feedback message type, AI feedback support type, AI feedback modality). Finally, Model 4 added methodological moderators (study design, SRL measurement method) to control for potential observational and study-level variance. The detailed regression coefficients, variance components, and stepwise model comparisons are summarized in Table 16 (Supplementary Material I).

The unconditional model (Model 0) indicated a significant overall positive effect of AI feedback on SRL (Intercept = 0.509,  $p < .001$ ). When contextual moderators were entered in Model 1, the model fit did not significantly improve (LRT Chi-square (8) = 9.01,  $p > .05$ ), explaining only 1.3% of the between-study variance. Within this step, an intervention length of 6–10 weeks showed a marginally significant positive effect compared to a single session.

The introduction of agent-related moderators in Model 2 resulted in a significant improvement in model fit (LRT Chi-square (4) = 13.61,  $p < .01$ ) and explained a substantial proportion of the between-study variance ( $R^2_{within} = 15.7\%$ ). Among these variables, feedback source emerged as a robust, statistically significant moderator: studies employing human-AI hybrid feedback reported significantly smaller SRL effects than those using AI-led feedback ( $-0.351$ ,  $p < .05$ ), a pattern that remained stable across all subsequent models. Additionally, the positive effect of the 6–10 weeks intervention length reached statistical significance in this model ( $0.331$ ,  $p < .05$ ).

When message- and implementation-related moderators were added in Model 3, no significant further improvement in model fit was observed (LRT Chi-square (9) = 7.35,  $p > .05$ ). Although several feedback features, including directive support, multichannel delivery, and the combination of metacognitive and motivational feedback, showed positive coefficients, none reached statistical significance after controlling for contextual and agent-related moderators.

Finally, the inclusion of methodological moderators in Model 4 also did not yield a significant improvement in model fit (LRT Chi-square (2) = 0.83,  $p > .05$ ). Across all models, a substantial proportion of within-study variance ( $I^2_{within(L2)} = 54.01\%$ ) and between-study variance ( $I^2_{between(L3)} = 35.73\%$ ) remained unexplained, underscoring the complexity of SRL as an outcome. Overall, the hierarchical meta-regression results indicate that the inclusion of agent-related moderators significantly improved model fit and accounted for a larger proportion of the between-study variance, with feedback source emerging as a statistically significant moderator. In contrast, contextual, message- and implementation-related, and methodological moderators did not yield significant incremental improvements in model fit after these moderators were taken into account. At the same time, the substantial residual variance suggests that additional influences not captured in the present models are likely to be involved.

### Within-Category Moderator Analysis for AI Technology Type on Overall SRL

The hierarchical meta-regression analysis (see Sect. 3.4) revealed that the inclusion of agent-related moderators in Model 2 significantly improved model fit ( $p < .001$ ),

with generative AI dialogue systems yielding significantly larger effects than other technologies. To further unpack these findings and explore how AI feedback distinctively influences SRL within specific technological contexts, a within-category moderator analysis was conducted (results see Table 17–110 in Supplementary Material I). This granular analysis examines whether the effectiveness of each AI technology type is contingent upon research characteristics.

For generative AI dialogue systems ( $k=154$ ), the effects were notably robust across most features. None of the moderators significantly altered the outcomes. However, a marginal effect was observed for feedback modality ( $F(1, 152)=3.63$ ,  $p=.059$ ). Multichannel feedback yielded a substantially larger effect ( $g=1.13$ ) compared to written-text only feedback ( $g=0.53$ ). Additionally, the feedback source showed a trend ( $p=.084$ ) where AI-led feedback ( $g=0.68$ ) outperformed human-AI hybrid approaches ( $g=0.24$ ). It should be noted that the multichannel subgroup included a limited number of studies ( $k=9$ ), warranting cautious interpretation. In contrast, the effectiveness of non-generative AI dialogue systems ( $k=53$ ) was heavily dependent on the SRL measurement method ( $F(1, 51)=11.48$ ,  $p=.001$ ). Studies utilizing behavioral traces reported a large effect size ( $g=1.30$ ,  $k=5$ ), whereas those relying on self-reports showed only a small effect ( $g=0.30$ ,  $k=48$ ). Similar to GenAI dialogue systems, a marginal advantage ( $p=.055$ ) was found for multichannel modalities ( $g=0.65$ ) over text-only formats ( $g=0.30$ ), reinforcing the potential value of multimodal interaction across different AI generations. For learning analytics systems ( $k=119$ ), study design emerged as a significant moderator ( $F(1, 117)=3.94$ ,  $p=.049$ ). RCTs demonstrated significantly stronger effects ( $g=0.48$ ) than QEDs ( $g=0.20$ ), suggesting that LAS interventions may require highly controlled environments to be effective. Finally, intelligent tutoring systems ( $k=61$ ) displayed high stability, with no significant differences found across any examined moderators (all  $p > .05$ ), indicating a standardized impact of ITSs on SRL.

## Publication Bias

Publication bias was assessed using a comprehensive approach that combined visual inspection with statistical tests adapted to the multilevel data structure. First, we examined contour-enhanced funnel plots (Peters et al., 2008) rather than standard funnel plots. Visual inspection revealed some asymmetry (see Fig. J2 in Supplementary Material J); however, the asymmetrical data points predominantly fell within the regions of statistical significance ( $p < .05$ ). This pattern suggests that the observed asymmetry is likely driven by between-study heterogeneity (e.g., small-study effects) rather than publication bias. Second, to rigorously test for bias while accounting for the dependency of effect sizes within studies, a multilevel meta-regression was conducted, using standard error as a moderator as a modified Egger's test (Rodgers & Pustejovsky, 2021; Schindler & Richter, 2023). The results indicated no significant risk of publication bias ( $t(385)=0.50$ ,  $p=.617$ ). Additionally, the trim-and-fill method (Duval & Tweedie, 2000) was applied as a sensitivity analysis. This analysis suggested that no missing studies were needed to restore symmetry, and the adjusted effect size remained identical to the original estimate ( $g=0.41$ , 95% CI [0.35, 0.48],  $p$

$< .001$ ). Collectively, these results suggest that publication bias was not a substantial threat to the validity of our findings.

## Quality Assessment

Quality assessment indicated that all studies exceeded the 0.60 quality threshold, with 77.7% meeting Kmet et al.'s (2004) more conservative 0.75 threshold, suggesting low risk of bias. All studies clearly articulated their research objectives and employed appropriate designs, including randomized controlled trials and quasi-experimental approaches. Neither investigator nor subject blinding was implemented in any study, likely due to practical constraints inherent to educational technology interventions. All studies utilized validated assessment tools and comprehensively reported participants' demographic characteristics (including academic level and age); however, few studies reported data on computer literacy and feedback literacy. While all studies detailed quantitative outcomes with data-substantiated conclusions, only 45.9% explicitly controlled for confounding variables. Nevertheless, given that only a small minority (8.2%) entirely neglected this measure, the findings suggest a need for refining control strategies rather than a fundamental lack of methodological awareness. The complete quality assessment, including all items and results, is detailed in Supplementary Material F (Tables F1 and F2).

## Discussion

### What are the Overall and Dimension-Specific Effects of AI Feedback on Students' SRL in Higher Education? (RQ1)

This meta-analysis provides robust evidence that AI feedback exerts a statistically significant, medium-sized positive effect on students' SRL in higher education. Moreover, sensitivity analyses restricted to RCTs yielded nearly identical estimates, lending confidence to the robustness and internal validity of the observed overall effect. This finding is directionally consistent with prior meta-analyses examining AI- or SRL-related interventions (e.g., Guan et al., 2024; Guo, 2022; Heung & Chiu, 2025; J. Wang & Fan, 2025), while suggesting larger effects relative to those reported in prior meta-analyses of non-AI feedback or SRL interventions (Cai et al., 2023; Panadero et al., 2023; Zheng et al., 2025).

Dimension-specific analyses reveal an asymmetrical pattern: strongest effects on the motivational and cognitive dimensions, and comparatively weaker effects on the behavioral and metacognitive dimensions. Through the lens of the conceptual framework, AI feedback may initially be perceived as an autonomy-supportive resource, potentially activating motivation and facilitating cognitive strategies (Wang et al., 2026; Xia et al., 2022). However, complex metacognitive reflection and sustained behavioral execution may place relatively higher cognitive demands on students (S. Liu et al., 2025; Ouyang 2025). In the absence of explicit scaffolding to help manage these demands, students' perceived coping potential may diminish (F. Wang et al., 2025), which may make some students less likely to remain on a growth pathway and

more likely to adopt defensive responses centered on effort reduction (Stadler et al., 2024). This pattern may help account for the comparatively weaker effects observed in the metacognitive and behavioral dimensions.

The wide 95% prediction interval [-0.661, 1.678] and the highly significant  $Q$ -statistic (2951.91) suggest substantial heterogeneity in the effects of AI feedback. Rather than indicating uniformly positive effects, the results point to meaningful variability in effect sizes across different study conditions. While supportive AI scaffolding can foster gains in cognitive elaboration, motivation, and behavioral dimensions of SRL (Azevedo et al., 2016; Dever et al., 2023; Xu et al., 2025), poorly aligned or insufficiently actionable feedback may be associated with reduced SRL effectiveness, which may manifest in patterns such as reduced engagement or increased anxiety (Aydin Yildiz, 2023; Chen, 2024; De Vetten, 2024). Beyond these immediate negative responses, backfire effects may also arise when AI substitutes for human agency. Under certain boundary conditions, students may default to metacognitive laziness, relying on cognitive offloading and passive adherence to automated guidance rather than engaging their own regulatory processes (Fan et al., 2024; Han et al., 2025). Taken together, these findings suggest that AI feedback may function as a double-edged resource, with its effects contingent on specific boundary conditions rather than being inherently beneficial.

## How Do Research Characteristics Moderate the Effects? (RQ2)

To unpack the heterogeneity in AI feedback effects on SRL, moderator analyses, particularly hierarchical meta-regression, indicate that agent-related moderators account for a substantial share of the observed variance. This pattern is broadly consistent with the proposed framework and suggests that how students initially appraise the feedback source may be a key factor of its effectiveness. Specifically, agent-related moderators, such as feedback source and AI technology type, appear to shape students' first appraisals of AI feedback, including perceptions of credibility, trust, and social presence. When these initial appraisals are perceived as threatening or insufficiently supportive, they may carry over into later evaluations of the feedback's clarity, usefulness, and controllability, thereby reducing students' willingness or ability to act on it. In this sense, the initial appraisal may set the tone for subsequent coping-related judgments and downstream regulation. Message- and implementation-related moderators, including feedback message type, modality, and support type, may further refine these later evaluations. Context-related moderators qualify the conditions under which these effects emerge, while methodological moderators account for variation in how they are estimated across studies. The following sections examine each moderator in more detail.

### 1. Feedback Source.

Moderator analyses identified feedback source as a significant moderator; however, contrary to hypothesis 1, AI-led feedback consistently demonstrated stronger effects than human-AI hybrid formats. While the direction of this effect was not anticipated, the salience of the feedback source is broadly consistent with the framework's

emphasis on agent-related moderators. Although human involvement was expected to support relatedness, the findings point to a more nuanced pattern. Hybrid designs have been associated with informational friction, often linked to limited synchronization between human and AI agents (Henderson et al., 2025; Xu & Cho, 2025). Instead of consistently yielding complementary support, such friction may be experienced as a potential ego threat that constrains autonomy and relatedness, and may in turn be associated with a shift toward a well-being pathway (Köpeczi-Bócz, 2025). In contrast, well-designed AI-led systems tend to provide a consistent, non-judgmental environment (Huang et al., 2025). By potentially minimizing the social-evaluative threats often inherent in human intervention (Kluger & DeNisi, 1996), AI-led feedback is more likely to foster the psychological safety necessary to sustain the growth pathway (Betson et al., 2025; Kaliisa et al., 2026). Practically, these findings caution against the mere juxtaposition of human and AI components. For instance, the role of AI can be defined as providing immediate, non-judgmental formative scaffolding, whereas the instructor can offer delayed socio-emotional and summative support. Such differentiation helps minimize conflicting cues and reduces threat appraisals (Banihashem et al., 2025b; Dang et al. 2025).

## 2) AI Technology Type.

Supporting hypothesis 2, generative AI dialogue systems and ITSs yielded stronger effects than non-generative AI dialogue systems and learning analytics systems. To understand this divergence, the systems can be examined through the lens of the conceptual framework. The limited effectiveness of static learning analytics systems implies that their abstract, outcome-oriented data are likely appraised as evaluative threats rather than support (Alfredo et al., 2024). Feeling overwhelmed or judged, students might prioritize protecting their self-esteem (the well-being pathway) over active learning or regulation (Boekaerts, 2011). This fragility is further reflected in within-category analyses, where the effectiveness of LASs was significantly moderated by study design. This pattern suggests that LASs may not operate as self-contained interventions that perform consistently across conditions. Instead, their effectiveness appears associated with the quality of instructional integration and the degree of environmental control, which may shape the extent to which students engage with the data rather than bypass it.

In contrast, the conversational nature of GenAI and the step-by-step guidance of ITSs offer a distinct user experience. By simulating a supportive partnership or social presence, these systems may facilitate the satisfaction of students' psychological need for relatedness (X. Zhao et al., 2025). Consequently, students are more likely to perceive the AI as a collaborator rather than a critic, fostering a commitment to the growth pathway. Because they inherently structure the regulatory process, GenAI and ITSs demonstrated remarkable robustness across various research characteristics, providing a stable psychological buffer for students. Practically, these findings suggest prioritizing dialogic, agent-like AI systems over static reporting tools (Bauer et al., 2025). Designers should leverage multimodal affordances not simply for engagement, but as essential scaffolds to reduce interpretive burden, ensuring

students sustain the competence necessary for complex regulatory processes (Yuan et al., 2026).

### 3) AI Feedback Message Type.

Although the overall moderation analysis did not reach statistical significance, descriptive comparisons revealed that combinatorial feedback (integrating cognitive, metacognitive, and motivational elements) consistently yielded larger effect sizes than single-component cognitive feedback. This statistical hierarchy aligns with the hierarchical meta-regression and conceptual framework's: agent characteristics function as a primary filter for psychological safety (Chan & Hu, 2023). If the AI is initially appraised as a threat, students may default to a defensive well-being pathway, rendering specific message content less impactful (Mesghina et al., 2025). Once this safety threshold is met, comprehensive combinations provide essential affective buffering. While isolated cognitive feedback primarily highlights performance discrepancies, which may risk competence frustration, combined feedback bolsters perceived resources by embedding cognitive corrections within motivational and strategic guidance. This framing empowers students to close performance gaps, encouraging sustained engagement in the growth pathway (Kunyan & Miao, 2025). From a design perspective, these findings support integrated architectures that embed metacognitive prompts within motivational framing over isolated cognitive feedback (Gavarkovs et al., 2023).

### 4) AI Feedback Support Type.

Although the overall moderation analysis for feedback support type was not statistically significant for overall SRL, the metacognitive subgroup analysis revealed a meaningful divergence: scaffolded and directive feedback yielded positive effects, whereas corrective feedback exhibited a negative trend, albeit based on a limited number of effect sizes. Viewed through the conceptual framework, this divergence highlights the critical role of perceived coping resources in shaping students' regulatory responses. Simple corrective feedback appears to signal a performance discrepancy without providing sufficient resources to close the gap. This perceived lack of coping potential likely frustrates the learner's need for competence, potentially triggering a shift toward the defensive well-being pathway where students disengage to protect self-esteem (Alsaïari et al., 2025). Interestingly, the unexpected positive effect of directive feedback suggests a beneficial mechanism of cognitive offloading (Y. Guo & Ye, 2026). Rather than inevitably inducing surface-level reliance, directive support may function as a necessary regulatory buffer that resolves immediate hurdles. By helping to secure the learner's perceived coping potential, it appears to preserve mental resources for higher-order metacognitive monitoring (Liu et al., 2023). While it may prevent immediate cognitive overwhelm, prolonged reliance on explicit directives carries the risk of unintended consequences, such as metacognitive laziness (Fan et al., 2024). In such scenarios, students may fail to internalize regulatory competencies, instead relying on AI-driven regulation as a substitute for their own self-regulatory processes. Practically, this highlights the importance of AI

systems going beyond static verification to provide adaptive support. System designers are encouraged to dynamically adjust the level of scaffolding, offering directive support to prevent initial disengagement when coping potential is low, and gradually transitioning to elaborative scaffolding to foster genuine regulatory autonomy.

### 5) AI Feedback Modality.

Feedback modality emerged as a marginally significant moderator for overall SRL, with multichannel feedback consistently outperforming written-text only formats. Notably, this multimodal advantage was particularly evident within the cognitive dimension. Furthermore, our subgroup data reveal that multichannel delivery provides a consistent advantage across both generative and non-generative AI dialogue systems. Interpreting this marginally significant trend through the lens of the conceptual framework, the consistent multimodal advantage can be theoretically elucidated by linking it to principles of cognitive load. Written-text-only feedback, particularly when it conveys complex regulatory guidance, can impose high decoding demands (Cosentino et al., 2025; Van Merriënboer & Sweller, 2010). Within the conceptual framework, this elevated cognitive burden may severely diminish students' secondary appraisal of their coping potential. In contrast, multichannel delivery synchronizes visual and verbal channels, generating processing fluency (Hwang, 2025). This multimodality likely helps by reducing extraneous cognitive load (Sweller, 2024; Trypke et al., 2024), making the feedback easier to digest and leaving students with enough mental capacity to actually apply the regulatory advice (Chiu, 2024; Mehta et al., 2025). By functioning as an essential cognitive scaffold (Ding et al., 2025), multimodality helps students appraise the feedback as a manageable challenge rather than an overwhelming threat, thereby satisfying competence needs and sustaining the growth pathway. The use of text-based formats to convey complex AI guidance could be associated with unintended consequences, as students might experience overload and attribute the resulting decoding difficulty to their own lack of ability (Zhang, 2024). Such misattributions may be perceived as ego-threatening, potentially undermining perceived competence and, in some cases, contributing to defensive disengagement. Practically, rather than serving merely as sensory enrichment, multimodal affordances would best be structured as strategic cognitive scaffolds. Such multichannel delivery minimizes decoding costs, preserving students' cognitive resources for substantive regulatory processes (C. Q. Zhao et al., 2026).

### 6) Intervention Length.

Although the moderation analysis for intervention length did not reach statistical significance, descriptive comparisons and our hierarchical meta-regression consistently revealed a non-linear, inverted-U trajectory. Interventions lasting 6–10 weeks demonstrated a statistically significant advantage over single-session interventions, whereas effects appeared to diminish for interventions exceeding 10 weeks. Viewed through the conceptual framework, short-term interventions (under 6 weeks) likely rely on a transient novelty effect, insufficient for fully satisfying relatedness needs or internalizing regulatory strategies. Conversely, the 6–10 week window provides ade-

quate time for students to establish psychological safety and solidify new regulatory habits (Guntur & Purnomo, 2024). Importantly, the declining effectiveness of longer interventions highlights a critical temporal constraint regarding learner autonomy. Prolonged, uninterrupted access may lead to unintended backfire effects, as students develop passive reliance and gradually outsource their regulatory responsibilities to the AI (Y. Zhao et al., 2025). This prolonged dependence likely frustrates the ultimate goal of autonomy, prompting a return to surface-level compliance. Practically, this temporal dynamic suggests that AI support would best be conceptualized as faded scaffolding (Bastani et al., 2025). Educators are encouraged to implement robust support during the initial habituation window, then gradually withdraw it to foster independent regulation (Song & Kim, 2021).

## 7) Academic Subject.

Contrary to hypothesis, academic subject did not significantly moderate the overall SRL effect, though a marginal advantage emerged for well-structured domains (e.g., natural sciences) in the metacognitive dimension. This divergence highlights a contextual boundary condition: in well-structured disciplines, AI generates precise, rule-based feedback that satisfies students' competence needs by serving as an objective calibration standard. Conversely, the diagnostic vagueness inherent to ill-structured domains risks frustrating students' perceived coping potential, inadvertently hindering effective metacognitive calibration (Singh et al., 2025).

## 8) Study Design.

Contrary to the hypothesis, study design did not significantly moderate the overall SRL effect. However, a significant divergence emerged specifically within the behavioral dimension and learning analytics systems, where RCTs unexpectedly demonstrated stronger effects than QEDs. While QEDs often inflate effect sizes due to selection bias (Li et al., 2023), this counterintuitive trend highlights the extreme sensitivity of behavioral execution to experimental conditions. Translating high-inference AI feedback into tangible behavioral regulation appears to heavily rely on the high-fidelity instructional integration characteristic of RCTs. In less controlled settings (de Vreugd et al., 2024), the regulatory process risks stalling (F. J.-Y. Jin et al., 2025), leaving students to engage superficially without converting their psychological appraisals into actual behavioral execution.

## 9) SRL Measurement Method.

Contrary to hypothesis 9, the SRL measurement method did not significantly moderate the effect of AI feedback on SRL. This indicates that AI feedback impacts both students' perceptions and observable behaviors with a similar magnitude. This finding aligns with recent methodological validation studies in SRL research. Rovers et al. (2019) showed that self-report and behavioral metrics aligned on global SRL but diverged at the fine-grained level, highlighting context sensitivity. Cheng et al.

(2025) and Zhang et al. (2021) further illustrated how network and clustering analysis of trace data can reliably reflect SRL behaviors observed through other methods.

## Limitations and Open Questions

Notwithstanding the substantive evidence presented, several limitations necessitate a cautious interpretation of these findings and highlight critical directions for future research. First, while agent-related moderators such as feedback source and AI technology type accounted for a portion of the observed variance, the current dataset precludes a comprehensive examination of how student-related moderators, notably AI literacy and baseline SRL, interact with feedback reception (Kaliisa et al., 2026). Second, with respect to domain specificity, AI feedback was found to yield stronger metacognitive gains in structured disciplines (e.g., natural sciences and engineering) than in less structured fields (e.g., humanities and social sciences). This suggests that current AI systems may be less effective at supporting metacognitive processes in domains where clear evaluative standards are lacking. However, due to limited available data, it remains unclear whether this difference reflects intrinsic technological constraints or discipline-specific prompt and implementation designs. Finally, although this study integrates self-determination theory and the dual-process model to construct a conceptual framework for explaining how research characteristics may modulate SRL, the meta-analytic nature of the data, particularly the use of aggregated effect sizes, does not allow for direct testing of the underlying psychological mechanisms, including primary and secondary appraisal. As such, the present findings offer only indirect support for these proposed mechanisms. The framework should therefore be understood as a theory-informed interpretive account rather than a fully validated process model. Future research would benefit from adopting process-sensitive methodologies, such as think-aloud protocols or neurophysiological measures, to provide more direct evidence of the appraisal processes associated with AI feedback.

## Conclusion and Educational Implications

Synthesizing 387 effect sizes from 85 studies, this meta-analysis confirms a significant, medium-sized positive effect of AI feedback on students' SRL in higher education. Dimension-specific analyses reveal an asymmetrical pattern: the facilitative impact is strongest for the motivational and cognitive dimensions, while effects on behavioral and metacognitive outcomes are comparatively weaker. Crucially, the wide 95% prediction interval, which extends into negative values, indicates that AI feedback effects may vary substantially across different conditions and may even be negative under certain circumstances. Our hierarchical meta-regression identified agent-related characteristics, particularly feedback source and AI technology type, as the primary drivers of this heterogeneity. AI-led feedback yielded significantly larger effect sizes than human-AI hybrid formats. Furthermore, dialogic systems (e.g., GenAI, ITSs) consistently outperformed LASs and non-generative AI dialogue systems. Overall, these findings not only quantify the effects of AI feedback on SRL

but also delineate its boundary conditions. These findings indicate that the implementation of AI feedback may trigger backfire effects, manifesting as cognitive offloading, metacognitive laziness, or passive reliance when not coupled with deliberate pedagogical integration and adaptive system designs.

Based on these findings, concrete implications are proposed to balance benefits and risks for key stakeholders. For instructors, prioritizing dialogic AI systems over static reporting tools can reduce the interpretive burden that often elicits defensive reactions. When integrating human and AI feedback, their roles should be clearly differentiated: AI can provide immediate, non-judgmental formative scaffolding, whereas human feedback can be reserved for delayed, socio-emotional summative support, thereby reducing informational friction. In addition, the data point toward a 6–10 weeks window as a potentially optimal habituation period, suggesting that AI support is best conceptualized as faded scaffolding within this timeframe. Strong initial support functions as a cognitive safety net, but should be gradually withdrawn to foster independent regulation and avoid passive reliance. For designers, the findings imply several system-level requirements. Architectures should extend beyond static verification by incorporating adaptive scaffolding mechanisms. Providing directive support when students' coping potential is low can serve as cognitive offloading that prevents disengagement. Multimodal affordances may also function as strategic scaffolds that reduce decoding costs and mitigate ability-related misattribution. Embedding metacognitive prompts within motivational framing can further buffer affective threats. For researchers, several methodological and theoretical priorities emerge. Transparent reporting of prompt engineering is essential to isolate the effects of social presence. It is also necessary to move beyond the traditional dependency assumption by identifying thresholds that distinguish beneficial cognitive offloading from detrimental metacognitive laziness. Finally, triangulating subjective self-reports with objective interaction logs should become routine practice to capture the temporal dynamics of self-regulated learning.

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**Data Availability** All data and code associated with this study are publicly available via the Open Science Framework ([https://osf.io/xd6uq/overview?view\\_only=f75ab7c9719e4ff9847661cd1d1158f](https://osf.io/xd6uq/overview?view_only=f75ab7c9719e4ff9847661cd1d1158f)).

## Declarations

**Ethical Approval** Not applicable (this research does not involve human participants).

**Competing interests** The authors declare that they have no competing interests.

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